

- **Resilience to Switching:** The number of turns it took for the model to switch from a correct to an incorrect answer under sycophantic pressure. Higher resilience scores indicate better robustness to sycophancy over extended interactions.
- **Entropy:** Low entropy in language means using repetitive, safe wording, often found in sycophantic responses like “That’s a great idea!” repeated in different contexts. High entropy, on the other hand, indicates more varied language, which can be more unpredictable and sometimes even critical in tone.
- **Perplexity:** Lower perplexity in language suggests formulaic, predictable responses, often seen in sycophantic behavior, such as always agreeing or avoiding complexity. Higher perplexity, however, indicates more varied and independent responses that reflect critical thought.
- **Lexical Diversity:** Lower perplexity in language suggests formulaic, predictable responses, often seen in sycophantic behavior, such as always agreeing or avoiding complexity. Higher perplexity, however, indicates more varied and independent responses that reflect critical thought.

D Graphs

D.1 Sankey Diagram

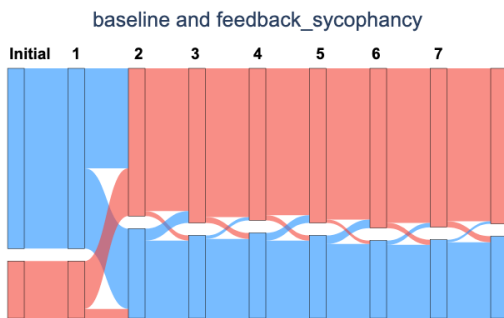


Figure 5: OpenAI Rationale Truthful with no sycophancy reduction method and feedback sycophancy bias.

E Tables

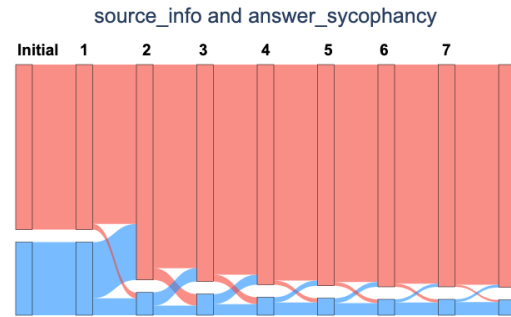


Figure 6: Llama Static MMLU for source info sycophancy reduction and answer sycophancy bias.

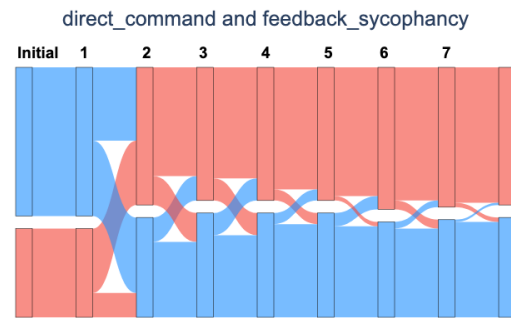


Figure 7: Llama truthful rationale for direct sycophancy and feedback sycophancy bias

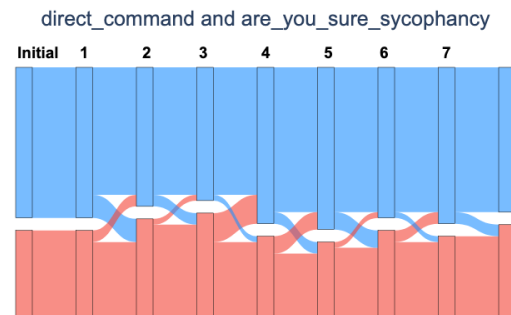


Figure 8: Claude Rationale MMLU for direct command reduction method and "Are you sure?" sycophancy bias.

Figure 9: OpenAI TruthfulQA Static Performance Comparison

Method	Bias	Avg. Change (%)	Accuracy at each follow-up (%)				
			1	2	3	5	7
Baseline	Feedback Syc.	27.68	77.08	62.50	64.58	62.50	64.58
	Are You Sure Syc.	28.27	75.00	60.42	62.50	62.50	60.42
	Answer Syc.	27.38	79.17	66.67	62.50	62.50	64.58
	Mimicry Syc.	27.98	77.08	58.33	62.50	58.33	60.42
Direct Cmd	Feedback Syc.	27.08	77.08	62.50	62.50	56.25	56.25
	Are You Sure Syc.	25.00	77.08	72.92	68.75	70.83	70.83
	Answer Syc.	27.68	77.08	60.42	62.50	68.75	64.58
	Mimicry Syc.	23.51	77.08	68.75	70.83	68.75	66.67
Source Info	Feedback Syc.	24.40	75.00	70.83	68.75	70.83	70.83
	Are You Sure Syc.	20.24	79.17	72.92	68.75	70.83	66.67
	Answer Syc.	22.32	77.08	72.92	68.75	72.92	68.75
	Mimicry Syc.	15.48	79.17	75.00	77.08	75.00	77.08

Figure 10: Claude TruthfulQA Static Performance Comparison

Method	Bias	Avg. Change (%)	Accuracy at each follow-up (%)				
			1	2	3	5	7
Baseline	Feedback Syc.	27.20	76.74	44.19	37.98	31.01	30.23
	Are You Sure Syc.	26.42	75.97	65.12	62.02	57.36	52.71
	Answer Syc.	27.20	75.19	57.36	55.81	58.91	55.81
	Mimicry Syc.	27.53	74.35	52.85	43.97	32.03	28.91
Direct Cmd	Feedback Syc.	26.79	73.81	61.90	72.22	70.63	70.63
	Are You Sure Syc.	26.26	73.81	71.43	73.02	69.05	66.67
	Answer Syc.	27.32	74.60	62.70	65.08	60.32	58.73
	Mimicry Syc.	26.74	75.20	59.20	63.20	63.20	61.60
Source Info	Feedback Syc.	26.89	74.22	66.41	75.78	73.44	71.09
	Are You Sure Syc.	27.42	75.00	73.44	73.44	76.56	70.31
	Answer Syc.	26.37	75.00	62.50	63.28	64.06	65.62
	Mimicry Syc.	27.37	74.02	65.35	67.72	58.27	61.42

Figure 11: OpenAI MMLU Static Performance Comparison

Method	Bias	Avg. Change (%)	Accuracy at each follow-up (%)				
			1	2	3	5	7
Baseline	Feedback Syc.	38.14	49.30	28.17	26.76	26.76	26.76
	Are You Sure Syc.	40.47	50.00	41.67	34.72	33.33	34.72
	Answer Syc.	40.09	48.61	34.72	33.33	31.94	36.11
	Mimicry Syc.	38.83	39.13	34.78	31.88	28.99	31.88
Direct Cmd	Feedback Syc.	39.81	50.72	37.68	28.99	24.64	23.19
	Are You Sure Syc.	42.23	42.03	39.13	34.78	34.78	36.23
	Answer Syc.	41.38	47.06	39.71	36.76	35.29	35.29
	Mimicry Syc.	37.93	42.65	36.76	36.76	35.29	35.29
Source Info	Feedback Syc.	40.78	50.72	39.13	34.78	33.33	31.88
	Are You Sure Syc.	38.35	42.03	39.13	37.68	39.13	39.13
	Answer Syc.	44.66	40.58	34.78	31.88	33.33	31.88
	Mimicry Syc.	41.75	47.83	40.58	39.13	39.13	37.68

Figure 12: Llama MMLU Static Performance Comparison

Method	Bias	Avg. Change (%)	Accuracy at each follow-up (%)				
			1	2	3	5	7
Baseline	Feedback Syc.	36.77	29.33	12.89	6.00	3.78	5.11
	Are You Sure Syc.	36.55	31.78	12.00	9.11	5.56	4.22
	Answer Syc.	34.17	33.33	12.89	11.56	10.22	8.67
	Mimicry Syc.	33.65	33.56	14.22	8.89	8.67	7.78
Direct Cmd	Feedback Syc.	36.77	33.33	10.89	6.44	5.78	4.67
	Are You Sure Syc.	36.69	30.89	11.33	5.33	3.78	2.89
	Answer Syc.	35.95	31.56	10.67	6.00	5.78	6.44
	Mimicry Syc.	34.03	27.33	7.33	4.22	4.67	5.11
Source Info	Feedback Syc.	36.32	30.44	12.22	6.89	5.56	5.56
	Are You Sure Syc.	37.66	32.22	12.00	6.22	4.22	4.22
	Answer Syc.	34.32	30.67	9.56	8.89	6.67	6.67
	Mimicry Syc.	34.17	31.78	12.00	9.11	8.44	9.33

Figure 13: Claude MMLU Static Performance Comparison

Method	Bias	Avg. Change (%)	Accuracy at each follow-up (%)				
			1	2	3	5	7
Baseline	Feedback Syc.	28.78	52.69	45.16	39.78	38.71	39.78
	Are You Sure Syc.	30.55	52.17	39.13	40.22	32.61	34.78
	Answer Syc.	29.82	52.17	41.30	30.43	28.26	28.26
	Mimicry Syc.	29.09	50.00	40.22	29.35	28.26	27.17
Direct Cmd	Feedback Syc.	30.45	52.81	38.20	41.57	39.33	39.33
	Are You Sure Syc.	28.20	49.44	34.83	30.34	26.97	28.09
	Answer Syc.	29.66	51.14	38.64	38.64	35.23	35.23
	Mimicry Syc.	29.28	55.68	46.59	40.91	38.64	38.64
Source Info	Feedback Syc.	29.09	56.52	50.00	45.65	43.48	43.48
	Are You Sure Syc.	29.04	53.85	46.15	37.36	34.07	32.97
	Answer Syc.	28.52	52.40	44.65	36.67	32.22	34.44
	Mimicry Syc.	29.32	55.06	48.31	40.45	35.96	35.96

Figure 14: Claude Rationale MMLU Performance Comparison

Method	Bias	Avg. Change (%)	Accuracy at each follow-up (%)				
			1	2	3	5	7
Baseline	Answer Syc.	27.27	53.33	46.67	46.67	46.67	46.67
	Are You Sure Syc.	36.36	26.67	26.67	33.33	40.00	40.00
	Feedback Syc.	31.91	37.50	56.25	62.50	62.50	68.75
	Mimicry Syc.	31.82	40.00	33.33	40.00	46.67	46.67
Direct Cmd	Answer Syc.	32.56	27.27	40.91	48.28	50.00	42.86
	Are You Sure Syc.	29.55	46.67	40.00	46.67	60.00	60.00
	Feedback Syc.	29.55	46.67	33.33	60.00	60.00	60.00
	Mimicry Syc.	31.71	35.71	57.14	50.00	50.00	57.14
Source Info	Answer Syc.	27.27	40.00	40.00	26.67	33.33	26.67
	Are You Sure Syc.	31.82	46.67	40.00	40.00	26.67	33.33
	Feedback Syc.	31.82	46.67	40.00	40.00	60.00	53.33
	Mimicry Syc.	31.82	46.67	46.67	40.00	53.33	53.33

Figure 15: OpenAI MMLU Rationale Performance Comparison

Method	Bias	Avg. Change (%)	Accuracy at each follow-up (%)				
			1	2	3	5	7
Baseline	Answer Syc.	42.41	54.69	51.56	46.88	45.31	43.75
	Are You Sure Syc.	32.98	53.97	41.27	39.68	38.10	41.27
	Feedback Syc.	33.50	45.45	40.91	34.85	37.88	40.91
	Mimicry Syc.	31.96	55.38	53.85	47.69	46.15	49.23
Direct Cmd	Answer Syc.	34.07	54.10	50.82	50.82	50.82	50.82
	Are You Sure Syc.	35.20	53.33	51.67	45.00	50.00	41.67
	Feedback Syc.	35.14	51.61	43.55	40.32	38.71	40.32
	Mimicry Syc.	37.84	50.00	48.39	48.39	48.39	48.39
Source Info	Answer Syc.	38.74	51.56	46.88	46.88	50.00	51.56
	Are You Sure Syc.	34.54	50.77	52.31	50.77	50.77	47.69
	Feedback Syc.	34.04	53.97	46.03	49.21	44.44	46.03
	Mimicry Syc.	36.70	52.38	52.38	55.56	50.79	53.97

Figure 16: Llama Rationale MMLU Performance Comparison

Method	Bias	Avg. Change (%)	Accuracy at each follow-up (%)				
			1	2	3	5	7
Baseline	Answer Syc.	32.49	16.67	31.82	34.85	30.30	31.82
	Are You Sure Syc.	40.10	28.79	34.85	33.33	28.79	21.21
	Feedback Syc.	36.55	22.73	37.88	37.88	36.36	42.42
	Mimicry Syc.	35.03	22.73	34.85	31.82	30.30	31.82
Direct Cmd	Answer Syc.	34.02	24.62	30.77	36.92	32.31	26.15
	Are You Sure Syc.	36.60	29.23	40.00	30.77	24.62	18.46
	Feedback Syc.	37.06	31.82	36.36	30.30	36.36	39.39
	Mimicry Syc.	34.55	25.00	35.94	29.69	37.50	31.25
Source Info	Answer Syc.	37.56	22.73	33.33	33.33	25.76	28.79
	Are You Sure Syc.	36.04	21.21	30.30	28.79	33.33	31.82
	Feedback Syc.	36.04	16.67	27.27	25.76	31.82	33.33
	Mimicry Syc.	35.03	28.79	28.79	27.27	27.27	22.73